

RESEARCH ARTICLE | AUGUST 15 2025

Bandgap opening induced by electron localization in graphene antidot lattices

Hai-Wei Luo ; Chang-Chun He  ; Yu-Jun Zhao ; Xiao-Bao Yang  



J. Chem. Phys. 163, 074701 (2025)

<https://doi.org/10.1063/5.0280786>



Articles You May Be Interested In

Electron transport in quantum antidots made of four-terminal graphene ribbons

J. Appl. Phys. (November 2009)

Energy spectrums of bilayer triangular phosphorene quantum dots and antidots

AIP Advances (April 2017)

Theoretical study on electronic properties of MoS₂ antidot lattices

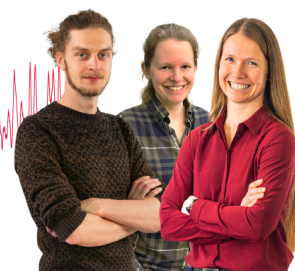
J. Appl. Phys. (September 2014)

Webinar From Noise to Knowledge

May 13th – Register now



Universität
Konstanz



Bandgap opening induced by electron localization in graphene antidot lattices

Cite as: J. Chem. Phys. 163, 074701 (2025); doi: 10.1063/5.0280786

Submitted: 14 May 2025 • Accepted: 27 July 2025 •

Published Online: 15 August 2025



Hai-Wei Luo,  Chang-Chun He,  Yu-Jun Zhao,  and Xiao-Bao Yang 

AFFILIATIONS

School of Physics and Optoelectronics, South China University of Technology, Guangzhou 510640, China

^{a)}Authors to whom correspondence should be addressed: scuthecc@scut.edu.cn and scxbyang@scut.edu.cn

ABSTRACT

Graphene antidot lattices (GALs) have garnered significant attention for their potential in semiconductor applications, yet the origin of bandgap opening remains controversial. Combining the octet rule, we propose a low-parameter physical model with weighted information entropy to quantitatively determine the electron density distribution, and the tight-binding parameters are obtained from the occupancy numbers based on the maximum entropy method. The results from our model reveal a complex bandgap opening mechanism in zigzag-edged hexagonal GALs (ZH-GALs), where specific inter-ribbon connections and quantum confinement cause the localization of π -electrons between antidots, leading to the elimination of energy levels degeneracy. We also observe that the anisotropy of rectangular ZH-GALs is enhanced as the defect radius increases, indicating a transition from GALs-like to graphene nanoribbons-like bandgap behavior. This study tells us that more than 1/9 ZH-GALs have considerable bandgaps, addressing the deficiency in band structure engineering between regimes dominated by defect scattering and quantum confinement.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0280786>

I. INTRODUCTION

Graphene, with its unique two-dimensional structure and excellent electronic, mechanical, and thermal properties,^{1–4} has garnered significant attention in the fields of condensed matter physics and materials science. Since the zero-gap character limits its application in semiconductor devices, modulating the bandgaps in graphene has become an important research topic. Considerable efforts^{5–13} have been made in developing graphene materials with semiconducting properties that exhibit excellent electronic transport properties. As a variant of quantum confined graphene, graphene nanoribbons (GNRs) can exhibit varying bandgaps depending on the edge configurations and ribbon widths.^{14–16} However, for electronic devices requiring large bandgaps in applications, it is hard for GNRs to carry sufficiently large currents. Another approach involves the periodic modification of graphene, such as B–N doping,¹⁷ H atom adsorption,⁵ and introduction of ring defects,¹⁸ which is essential for maintaining its exceptionally high charge carrier mobility. Creating periodic holes in pristine graphene to form graphene antidot lattices (GALs), also known as graphene nanomeshes (GNMs),¹⁰ have shown exceptional performance in field-effect transistors,^{7,13}

molecular sieving,¹⁹ and supercapacitors.²⁰ In recent years, significant progress has been made in the experimental synthesis of graphene nanomaterials. Jessen *et al.* utilized lithography to define a superlattice in graphene encapsulated in hexagonal boron nitride, achieving band structure engineering through quantum confinement effects.²¹ Their measurements confirm the theoretical predictions in sizable¹⁰ and magnetically tunable²² bandgap. A major breakthrough came in 2018 when Moreno *et al.* synthesized high-quality, large-area nanoporous graphene (NPG) by inter-ribbon coupled GNR, which can be considered a type of GAL, with atomic precision on an Au(111) surface.⁷ Starting from well-designed small precursors, the bottom-up on-surface synthesis strategies not only provided an unprecedented route for synthesizing complex and atomically precise low-dimensional carbon-based nanostructures, such as Janus GNRs,²³ with great success, but also have enabled the application of GALs in quantum devices,²⁴ as discussed in the recent review by Qin *et al.*²⁵

Numerous studies have investigated GALs with various defect shapes or unit cell shapes, which exhibit different bandgap characteristics.^{6,26–30} For graphene antidot lattices with zigzag-edged hexagonal holes (ZH-GALs), the bandgap is sensitive to the lattice

parameters because the balance between two sublattices is maintained. A widely accepted bandgap opening rule indicates that for all unit cell shapes of ZH-GALs, a large bandgap is observed only when $(m_1 - n_1) \bmod 3 = 0$ and $(m_2 - n_2) \bmod 3 = 0$,^{6,26} as shown in Fig. 1(a). This rule was subsequently explained based on defect scattering theory^{27,28} or the Clar sextet rule,^{29,30} enhancing the efficiency of subsequent electronic structure research. Consequently, it indicates that only 1/9 of the GALs are semiconductors.

In recent studies, Gregersen *et al.* observed that some structures exhibit unusual bandgap behavior that deviates from the typical bandgap opening rules.³¹ These structures have a certain value of bandgap, which may be attributed to the small spacing between holes. In addition, many experimental results and theoretical calculations indicated that the synthesized NPG possesses semiconductor properties and even exhibits tunable anisotropy.^{7,32–36} As shown in Fig. 1(a), in the tight-binding level bandgap calculation of ZH-GALs, there is a subset of structures that exhibit different behaviors depending on the hopping integrals used. The bandgap opening rules might be violated since the band structure provided by the nearest neighbor tight binding model (NNTB) may be

oversimplified. Thus, a more detailed understanding of the electronic properties of GALs is essential.

To achieve a more accurate understanding of the band structure of GALs, it is intuitive to correlate the bonding strength with the hopping integrals. The Kekulé structure depicts the electronic structure of simple sp^2 carbon materials with alternating single and double bonds. However, in complex electron-delocalized systems, the distribution of bonding strength is related to the bonding environment of electrons, which means that accurate prediction of C–C bond lengths is challenging. For the periodic structure shown in Fig. 1(b) with four Kekulé structures, the octet rule³⁷ cannot allocate weights between resonance structures, thus failing to determine the electron density distribution similar to the result from density functional theory (DFT) calculations in Fig. 1(c).

In this paper, we focus on the variation of the bandgap among several ZH-GALs with different unit cell shapes and defect radii. A low-parameter physical model was proposed to quantitatively determine the electron density distribution and energy bandgap of GALs. By conducting a brief classification of the bonding environment, we obtained accurate predictions of bond length distribution in GALs.

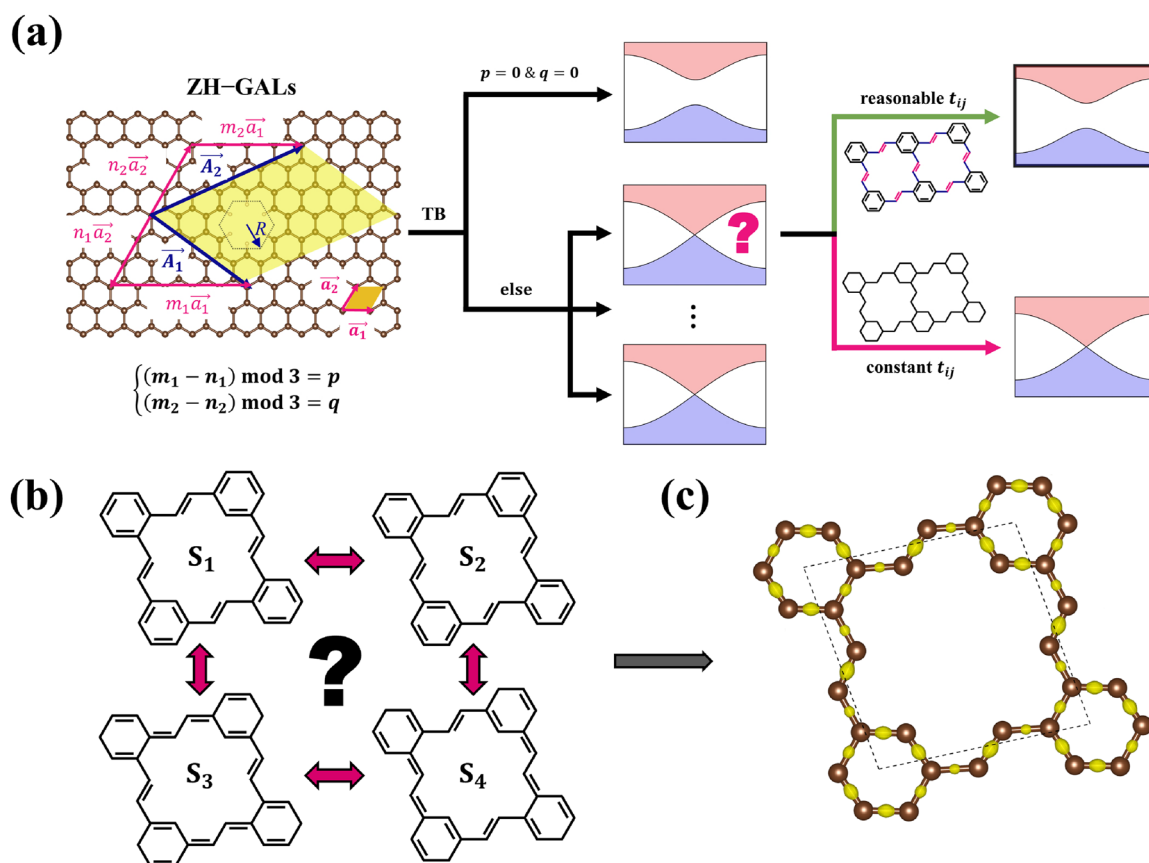


FIG. 1. (a) Classification method for semiconductor ZH-GALs. The left part shows the lattice structure of $\{[m_1, n_1], [m_2, n_2]\}$; the middle part illustrates the typical bandgap opening rules for ZH-GALs; and the right part shows the results obtained from TB calculations with different hopping integrals t_{ij} . (b) All four Kekulé structures of $\{[1, 2], [-3, 2]\}$. (c) Charge density difference of $\{[1, 2], [-3, 2]\}$ obtained from DFT calculations. The C atoms are denoted by the brown spheres, while H atoms are not drawn in the figure. The isosurface is set at 0.07 e/bohr³.

Adapting the occupancy numbers (ONs) to the parameterization of the tight-binding model, the maximum entropy tight-binding (METB) model demonstrates greater consistency with DFT in the details of the band structure. The bandgap opening is attributed to significant local interactions of π -electrons, and the bandgap variations are distinctly correlated with unit cell size changes with a period of three in rectangular ZH-GALs, which is less restrictive than the typical bandgap opening rule.

II. METHOD

A. Nomenclature

For GALs, the unit cell can be represented by a pair of lattice vectors $\{\mathbf{A}_1, \mathbf{A}_2\}$. As shown in Figure 1(a), $\mathbf{A}_1$ and $\mathbf{A}_2$ can be expressed as linear combinations of graphene lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$,

$$\begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix} = \begin{pmatrix} m_1 & n_1 \\ m_2 & n_2 \end{pmatrix} \begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{pmatrix}.$$

Thus, structures can be formally denoted by the set $\{[m_1, n_1], [m_2, n_2]\}$. To further classify GALs, the residues p or q are also employed to characterize the structures with the same $(m-n) \bmod 3$. For example, the structure depicted in Fig. 1(a) is characterized by the parameters $m_1 = 5$, $n_1 = -3$, $m_2 = 4$, and $n_2 = 3$. Consequently, this structure can be represented as $\{[5, -3], [4, 3]\}$, which belongs to $\{2, 1\}$.

In our study, all the antidots considered are passivated with hydrogen atoms at the edge. For hexagonal defects, we can directly determine the size of the defect by the number of removed carbon atoms N_{removed} , such as ZH₆-GALs shown in Fig. 1(a). In addition, the defect radius R in the figure is defined by $N_{\text{removed}} = 6R^2$ in subsequent studies. In this way, we are able to systematically represent GALs with various unit cell shapes and defect sizes.

B. Maximum entropy method

Due to the delocalization of π -electrons in carbon nanomaterials, the bonding strength distribution within different GALs presents various patterns. While the Clar structure offers insights into the bonding mechanisms in GALs, the precise quantification of electron density remains challenging. Recently, we proposed a statistical model,³⁸ combining the octet rule and maximum entropy principle without external parameters, successfully predicted the electron density distribution and total energy of PAHs, carbon nanoribbons, and boron systems³⁹ such as boranes, planar clusters, hollow cages, and bilayer clusters. It provides a comprehensive depiction of electron-delocalized systems with clear physical insight and quantitative accuracy. In this paper, we propose the maximum entropy method with the classification of bonding environments, which improves the accuracy of predicting electron density distribution for GALs.

As illustrated in Fig. 2(a), our model allocates valence electrons to C–C bonds under the octet rule. Here, N_{ele} denotes the total valence electrons in the unit cell, and n_i represents the electrons on the i th bond. The electron probability for the i th bond is given by $p_i = \frac{n_i}{N_{\text{ele}}}$. The occupancy numbers (ONs) for the i th bond is $\text{ON}_i = \frac{n_i}{2}$. Assuming uniform electron distribution over bonds, as

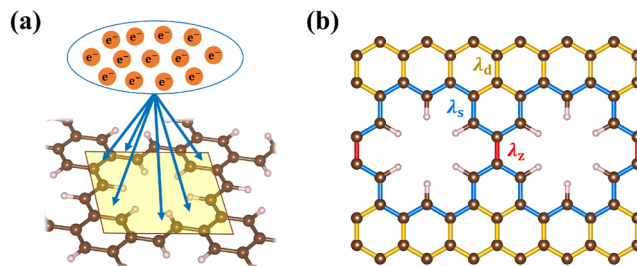


FIG. 2. (a) Total electron allocation model for $\{[1, 2], [-3, 2]\}$. (b) Schematic of the classification of local bonding environments between carbon atoms. The three types of C–C bonds are marked in yellow, blue, and red, respectively, based on the decreasing number of adjacent carbon hexagonal rings.

in graphene, we use Shannon entropy to measure electron distribution and apply the maximum entropy principle to obtain a reliable electron probability distribution.

Unlike graphene, hole defects in GALs introduce a variety of bonding environments. The bond lengths of C–C bonds located far from the defect tend to be uniform, which is significantly different from those at H-passivated edges, where the bond lengths deviate more from 1.42 Å. Therefore, the bonding environments of different configurations of GALs vary significantly. To achieve a more accurate prediction of the bonding strength distribution in all GALs, we classify the bonding environment based on the positional relationship between bonds and hole defects. As shown in Fig. 2(b), the bonding environments can be distinctly classified into three classes with a nearest-neighbor cutoff, which are denoted by yellow, blue, and red, indicating that they are shared by 2, 1, and 0 carbon hexagonal rings, respectively. After classification, the bonding entropy formula of the system can be written as the weighted sum of the Shannon entropy of the three types of bonds,

$$S_{\text{bond}} = -\lambda_d \sum_{d=1}^{N_d} p_d \ln p_d - \lambda_s \sum_{s=1}^{N_s} p_s \ln p_s - \lambda_z \sum_{z=1}^{N_z} p_z \ln p_z, \quad (1)$$

where λ_d , λ_s , and λ_z serve as the weight of these three types of bonding entropy. When surrounded by more hexagonal rings, the electrons in the vicinity should be less significantly influenced by the H-passivated edges, thereby possessing an increasing λ . N_d , N_s , and N_z represent the number of bonds in each category, respectively. It should be noted that the sum of N_d , N_s , and N_z equals the total number of bonds N_{bond} . As the absolute value of S_{bond} does not have a practical meaning, only two parameters need to be determined. In our work, the values used for the constants λ_d/λ_s and λ_z/λ_s were 1.058 and 0.837, respectively, which were obtained through the least-squares fitting of bond lengths from DFT calculations, as detailed in the [supplementary material](#). To assemble the training dataset, single holes were introduced in all graphene supercells, with volumes ranging from 8 to 24 graphene unit cells (ZH₆-GALs_{8–24}). In addition, to enhance the robustness of the model, 9 porous GALs were included, as detailed in the [supplementary material](#). For a given structure, the electron probabilities for all C–C bonds form a set $\{p_i\}$. With each carbon atom satisfying the octet rule, the electron probability distribution that maximizes S_{bond} represents the electron density distribution of the system.

For bandgap estimation, the nearest-neighbor tight-binding model is efficient for carbon nanomaterials, but can misjudge semiconductors and metals with constant hopping integrals. Aiming to improve the accuracy of the tight-binding model in predicting semiconducting properties for GALs, we propose the METB model that parameterizes the hopping integrals based on occupancy numbers, where the hopping integral between nearest neighbors is assumed to be proportional to the corresponding ON. Thereby, the Hamiltonian for π -electrons is given by

$$H_{\text{METB}} = - \sum_{\langle i,j \rangle} (\alpha \text{ON}_{ij} + \beta) (c_i^\dagger c_j + h.c.), \quad (2)$$

where i and j represent the indices of a pair of nearest-neighbor atoms, and ON_{ij} represents the occupancy number of the bond formed by these two atoms. In our work, the values used for the constants α and β are 0.70 and 1.37 eV, respectively, fitted via HOMO and LUMO data calculated on the PBE-GGA level. The implications of PBE underestimation for METB are discussed in the [supplementary material](#). Therefore, when studying different GALs, the maximum entropy method enables the integration of electronic probability distribution and band structure properties for analysis, thereby allowing for a more in-depth investigation into the physical mechanisms underlying the bandgap variations in GALs.

C. First-principle calculations

The calculations of GALs from training and test dataset were performed using the density functional theory (DFT) framework as implemented in VASP.^{40,41} The electron–electron interactions were described using the Perdew–Burke–Ernzerhof (PBE)⁴² parameterization of the generalized gradient approximation (GGA). All structures were treated as planar structures for the calculations. The plane-wave basis set with a kinetic energy cutoff of 520 eV was chosen based on convergence tests. The Brillouin zone is sampled with

a k-point spacing of $0.1 \text{ \AA}^{-1}$, utilizing a Γ -centered Monkhorst–Pack grid. All atomic positions are relaxed until the forces on each atom are less than $0.01 \text{ eV/\AA}$. A vacuum of $16 \text{ \AA}$ along the z axis is added to each plane to avoid the artificial interaction between graphene layers. A simplified k-path²⁸ is utilized to effectively characterize the band structure of the GALs.

III. RESULTS AND DISCUSSION

To investigate the correlation between electron density distribution and bandgap behavior of GALs, we first validated the maximum entropy method in GALs with small unit cell for reliability. Subsequently, we discussed the π -electron localization due to specific inter-ribbon connections at various defect rates. Finally, we expanded our study to larger rectangular GALs with increasing R to demonstrate the bandgap opening caused by π -electron localization.

A. Effects of the maximum entropy method on GALs

With the given bonding environment parameters, [Fig. 3\(a\)](#) illustrates the variation of bonding entropy with p_1 and p_2 using $\{[1, 2], [-3, 2]\}$ as an example. The triangular feasible region is determined by the octet rule. The three vertices indicate the distributions corresponding to the four Kekulé structures depicted in [Fig. 1\(b\)](#), which are associated with low S_{bond} . Herein, S_{best} represent the optimized solutions after the classification of bonding environments. Notably, compared to the corresponding p_1 of S_{best} , the unclassified result shows a significant increase in p_1 , causing incorrect bond length ordering in $\{[1, 2], [-3, 2]\}$, as shown in [Fig. S2](#) in the [supplementary material](#), which leads to an erroneous bandgap closure in subsequent METB calculations. This problem is resolved after classification, which also confirms that classifying the bonding environment using a nearest-neighbor cutoff has met our precision requirements.

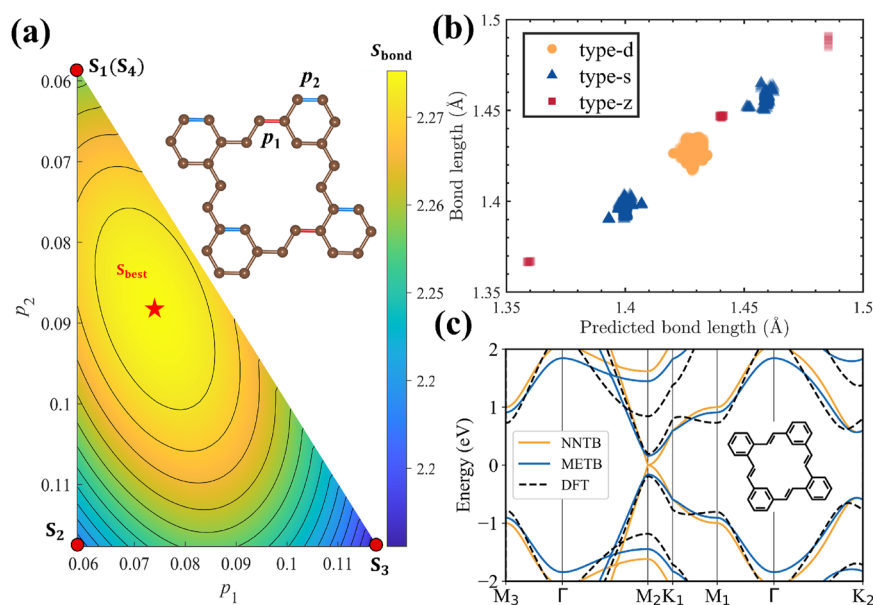


FIG. 3. (a) Bonding entropy surface S_{bond} varied with p_1 and p_2 for $\{[1, 2], [-3, 2]\}$. (b) Bond lengths predicted by the maximum entropy method and DFT. (c) Band structure of $\{[1, 2], [-3, 2]\}$.

To validate our model's robustness, we predict bond lengths for $\text{ZH}_6\text{-GAL}_{25-36}$ with the linear correlation between ON and bond length, as shown in Fig. 3(b). The bond length prediction results from the maximum entropy method show excellent agreement with DFT calculations, with only 0.41% of the predicted bond length deviate more than 0.01 Å. Interestingly, both the bond lengths predicted in the figure and the DFT results indicate a segmented distribution of bond lengths in $\text{ZH}_6\text{-GAL}_{25-36}$. We simply attribute this to the similar edges resulting from the identical defect shape. After coloring three types of C–C bond shown in Fig. 3(b), the data indicate that the bond lengths of type-d C–C bonds are typically around 1.42 Å. With fewer hexagonal rings adjacent to the C–C bonds, the bond length distribution tends to shift toward the extremes. This demonstrates that the bonding environment parameters effectively distinguish the behavior of electronic distribution as unit cell size increase. Given that the GALs in our training dataset all have unit cell volumes less than 25 times that of a graphene unit cell, the perfect prediction of bond lengths for $\text{ZH}_6\text{-GAL}_{25-36}$ suggests that the maximum entropy method can effectively describe the electron density distribution and bond lengths in larger GALs using only two independent bonding environment parameters.

It has been observed that NNTB sometimes provide incorrect bandgaps when the bond strength between different C–C bonds varies significantly, which typically corresponds to a more localized electron density distribution. The hopping integral of NNTB is set to $t_0 = 3.0$ eV. For the band structure of $\{[1, 2], [-3, 2]\}$ shown in Fig. 3(c), we find that NNTB indicates a bandgap closure at M_2 , while MMTB opens a bandgap of ~ 0.38 eV, consistent with DFT. This demonstrates the unreliability of NNTB in analyzing the electronic structure of ZH-GALs. To further demonstrate the effectiveness of METB in addressing GALs, we constructed 39 porous GALs. These structures were obtained by simple polycyclic aromatic hydrocarbons (PAH). Taking pyrene as an example, we have labeled all the

H-passivated C atoms in Fig. 4(a). Among these labeled sites, two pairs of carbon atoms are selected and respectively connected via two C atoms to form the porous GALs structure. As shown in Fig. 4(b), sites (2, 8) and (5, 10) from Fig. 4(a) are bridged to form porous GALs with unique defect shape. The data presented in Fig. 4(c) are derived from all 39 structures generated by benzene, naphthalene, and pyrene. The bandgap distribution of these porous GALs obtained by DFT is relatively continuous. METB roughly reproduces the calculation results of DFT, showing good consistency. The slight underestimation of the bandgap from METB may be attributed to the steric hindrance effect introduced by some H atoms. It is worth noting that the minimum bandgap value obtained by DFT calculation in these structures is ~ 0.075 eV. Unlike the above-mentioned results, nearly 50% of the structures are regarded as semimetals by NNTB. By combining the ON distribution with band structure, we can efficiently conduct an accurate analysis of the electronic structure of GALs.

In addition, we also conducted a brief discussion on disordered GALs, as detailed in the [supplementary material](#).

B. Bandgap opening of ribbon-shaped $\text{ZH}_6\text{-GALs}$

To obtain a more comprehensive understanding of the bandgap opening rules in zigzag-edged hexagonal graphene antidot lattices (ZH-GALs), it is necessary to consider ZH-GALs with various unit cell shapes, including rectangular, rhombic, and parallelogram-shaped structures. Therefore, we focus on $\text{ZH}_6\text{-GAL}_{8-36}$. Consistent with previous reports,^{6,26} structures satisfying $(m_1 - n_1) \bmod 3 = 0$ and $(m_2 - n_2) \bmod 3 = 0$ have larger bandgaps at the same defect concentration, as shown in Fig. S3 in [supplementary material](#). Their bandgaps follow the scaling law proposed by Pedersen *et al.* for semiconducting GALs,

$$E_g = a \frac{\sqrt{N_{\text{removed}}}}{N_{\text{total}}}, \quad (3)$$

where N_{removed} and N_{total} are the number of removed and original carbon atoms, respectively. However, the number of semiconductors in $\text{ZH}_6\text{-GAL}_{8-36}$ significantly deviates from the 1/9 predicted by the typical bandgap opening rule.

Figure 5(a) illustrates the bandgaps distribution of 158 structures obtained via DFT. The data show that only 17% of the structures satisfy the rule for bandgap opening, while 21% of the structures have a bandgap greater than 0.5 eV. In addition, structures with a bandgap less than 0.06 eV account for only 37% of these structures. The results above imply the existence of bandgap opening mechanisms in ZH-GALs that have not been revealed. For further study, we identified $\{[3, 0], A_2\}$ and $\{[2, 2], A_2\}$ from $\text{ZH}_6\text{-GAL}_{8-36}$ as the representative structures for studying electron distribution and bandgap behavior. The representative structures are listed in Table I.

For $\{[3, 0], A_2\}$ structures, depicted in the left panel of Fig. 5(b), these configurations consist of interconnected nanopores in graphene, with adjacent defects sharing two carbon atoms. These structures can be viewed as combinations of gulf-edged graphene nanoribbons (g-GNRs) connected via C–C bonds. The experimentally synthesized g-NPG,³⁴ reported by Piquero-Zulaica *et al.*, belongs to the $\{[3, 0], A_2\}$ class and exhibits a slightly smaller bandgap compared to its precursor g-GNR at the DFT level. In

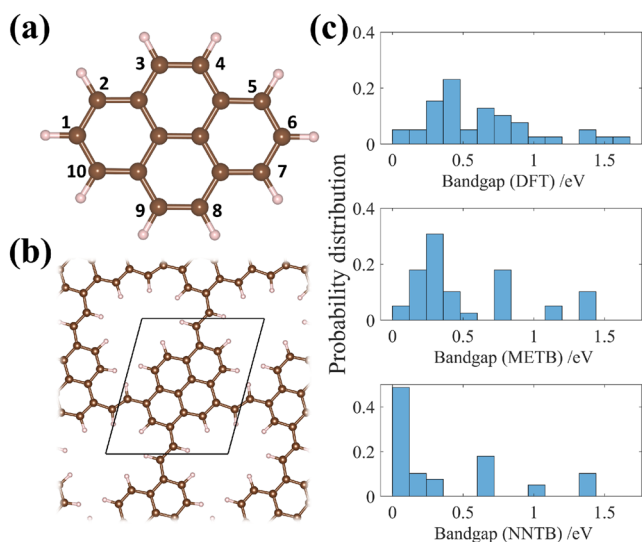


FIG. 4. (a) Schematic of pyrene. H-passivated carbon atoms are labeled. (b) Porous GAL constructed from pyrene. (c) The bandgap distribution of 39 porous GALs constructed from benzene, naphthalene, and pyrene.

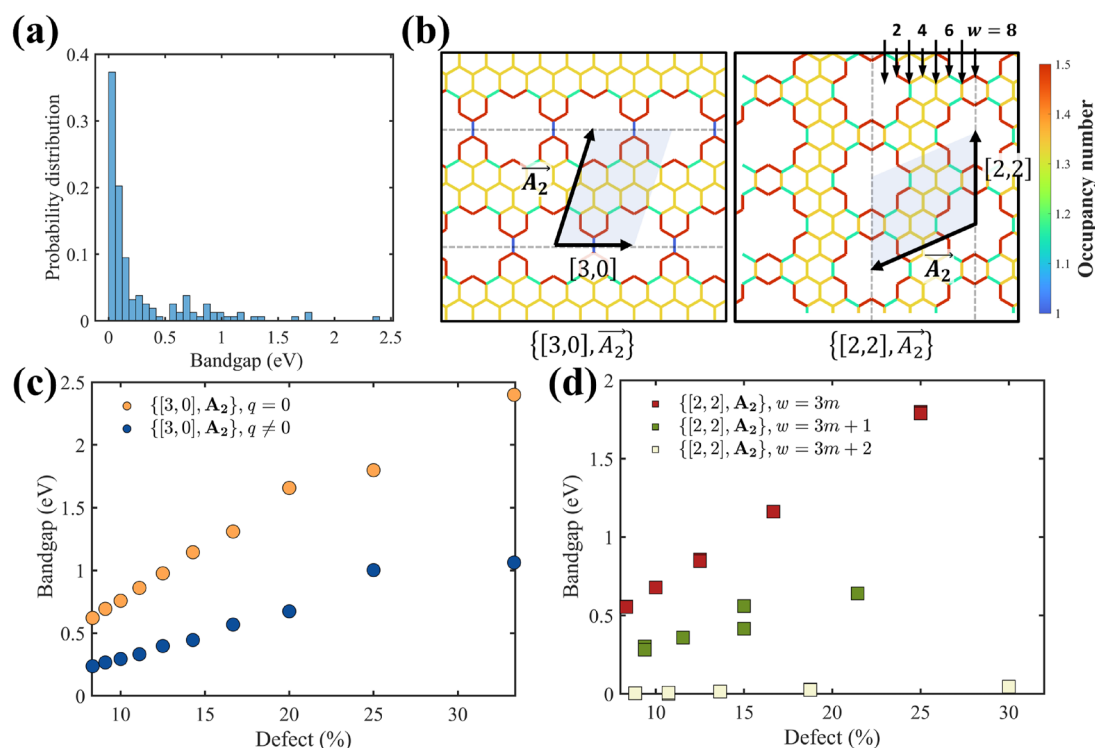


FIG. 5. (a) Bandgap distribution of ZH₆-GALs₈₋₃₆. Structures that satisfy the typical bandgap opening rule are indicated by the yellow dots. (b) Electron density distribution of $\{[3,0], A_2\}$ and $\{[2,2], A_2\}$. The width between two adjacent columns of defects is denoted by w . (c) Bandgap for $\{[3,0], A_2\}$ as a function of defect rate. Here, $q = (m_2 - n_2) \bmod 3$. (d) Bandgap for $\{[2,2], A_2\}$ as a function of defect rate. Here, m can be any positive integer.

TABLE I. Bandgaps for different cell types. Here, m_1, n_1, m_2, n_2 represent the shape of the unit cell $\{[m_1, n_1], [m_2, n_2]\}$, while N_{total} is the number of carbon atoms in the cell before defect formation.

Cell type	m_1	n_1	m_2	n_2	N_{total}	Defect%	Band gap (eV)
$\{[3,0], A_2\}$	3	0	-6	-12	72	8.3	0.62
	3	0	-7	-12	72	8.3	0.24
	3	0	-5	10	60	10.0	0.76
	3	0	-6	10	60	10.0	0.29
	3	0	-4	5	30	20.0	1.66
	3	0	-3	5	30	20.0	0.67
	3	0	0	3	18	33.3	2.40
	3	0	-1	3	18	33.3	1.06
$\{[2,2], A_2\}$	2	2	-6	5	44	13.6	0.01
	2	2	-5	5	40	15.0	0.56
	2	2	-5	4	36	16.7	1.16
	2	2	-4	4	32	18.8	0.03
	2	2	-4	3	28	21.4	0.64
	2	2	-4	2	24	25.0	1.80
	2	2	-2	3	20	30.0	0.05

Fig. 5(c), for $\{[3,0], A_2\}$ structures in ZH₆-GALs₈₋₃₆, the structure adheres to the bandgap opening rule only when $q = m_2 - n_2$ is divisible by three. However, both DFT and the METB calculations indicate a significant bandgap for all $\{[3,0], A_2\}$, rendering all these

configurations to follow Eq. (3) with two distinct slopes. The ONs on each C-C bond determined by the maximum entropy method are shown in Fig. 5(b). It is worth noting that the bond shared by two adjacent defects exhibit an occupancy number of 1. This suggests weak bonding strength and small hopping integrals, implying that the bandgap variation in $\{[3,0], A_2\}$ structures is primarily governed by the corresponding g-GNR. Recent studies on g-GNRs support that for g-GNRs with an edge hexagon distance of 3 and a gulf size of 2, the bandgaps exhibit variations following the patterns of Δ and $\frac{1}{2}\Delta$ as a function of nanoribbon width,¹⁵ which is consistent with the trends we identify in $\{[3,0], A_2\}$. The extra bandgap opening caused by the localization of π -electrons between defects is also observed in $\{[2,2], A_2\}$ and the subsequent ZH-GALs with larger R .

For $\{[2,2], A_2\}$ structures, depicted in the right panel of Fig. 5(b), these configurations feature multiple columns of defects aligned along the armchair direction; each defect was separated by two rows of carbon atoms in a column. Different from $\{[3,0], A_2\}$, the bandgap variation in $\{[2,2], A_2\}$ with increasing defect concentration exhibits three distinct trends, as shown in Fig. 5(d). To characterize the bandgap opening mechanism, we introduce a parameter w , analogous to the width of GNRs, to describe the width between two adjacent columns of defects. An inverse proportionality exists between the bandgap of $\{[2,2], A_2\}$ and w . The bandgaps for $\{[2,2], A_2\}$ can be described by the rule $\Delta_{3m} > \Delta_{3m+1} > \Delta_{3m+2} \neq 0$ for any m , consistent with METB calculations. This classification

is reminiscent of ac-GNRs but with a reversed bandgap order for $3m$ and $3m + 1$.¹⁶ Notably, DFT calculations yield tiny but non-zero Δ_{3m+2} , with the largest value being $\Delta_5 = 0.046$ eV, whereas NNTB predicts $\Delta_{3m+2} = 0$ for any m . Similarly, the distribution of ONs in Fig. 5(b) provides a clear explanation for the bonding strength of $\{[2, 2], A_2\}$. Unlike the delocalized regions marked in yellow, the alternating red and green ONs between adjacent defects indicate strong electron localization, as detailed in Fig. S4 in the [supplementary material](#). This also leads to a bandgap behavior similar to GNRs. To illustrate the specific inter-ribbon connections of $\{[2, 2], A_2\}$, we calculated the local density of states (LDOS) at the band edge in the [supplementary material](#).

C. Bandgap of ZH-GALs with larger R

For GALs with a large unit cell, the bandgap behavior can be interpreted through defect scattering^{27,28} when R is sufficiently small. However, when the defects are separated merely by atomic or molecular bridges, computational results for NPG reveal the semi-conducting properties, which is attributed to quantum confinement effects.^{7,33} To achieve a profound comprehension of the bandgap opening induced by quantum confinement, the maximum entropy method and METB were utilized to explore the π -electron localization and band structure of the system $\{[26, 0], [-10, 20]\}$ as R progressively increases. Since the rectangular lattice does not possess the high symmetry of a triangular lattice, this difference, combined

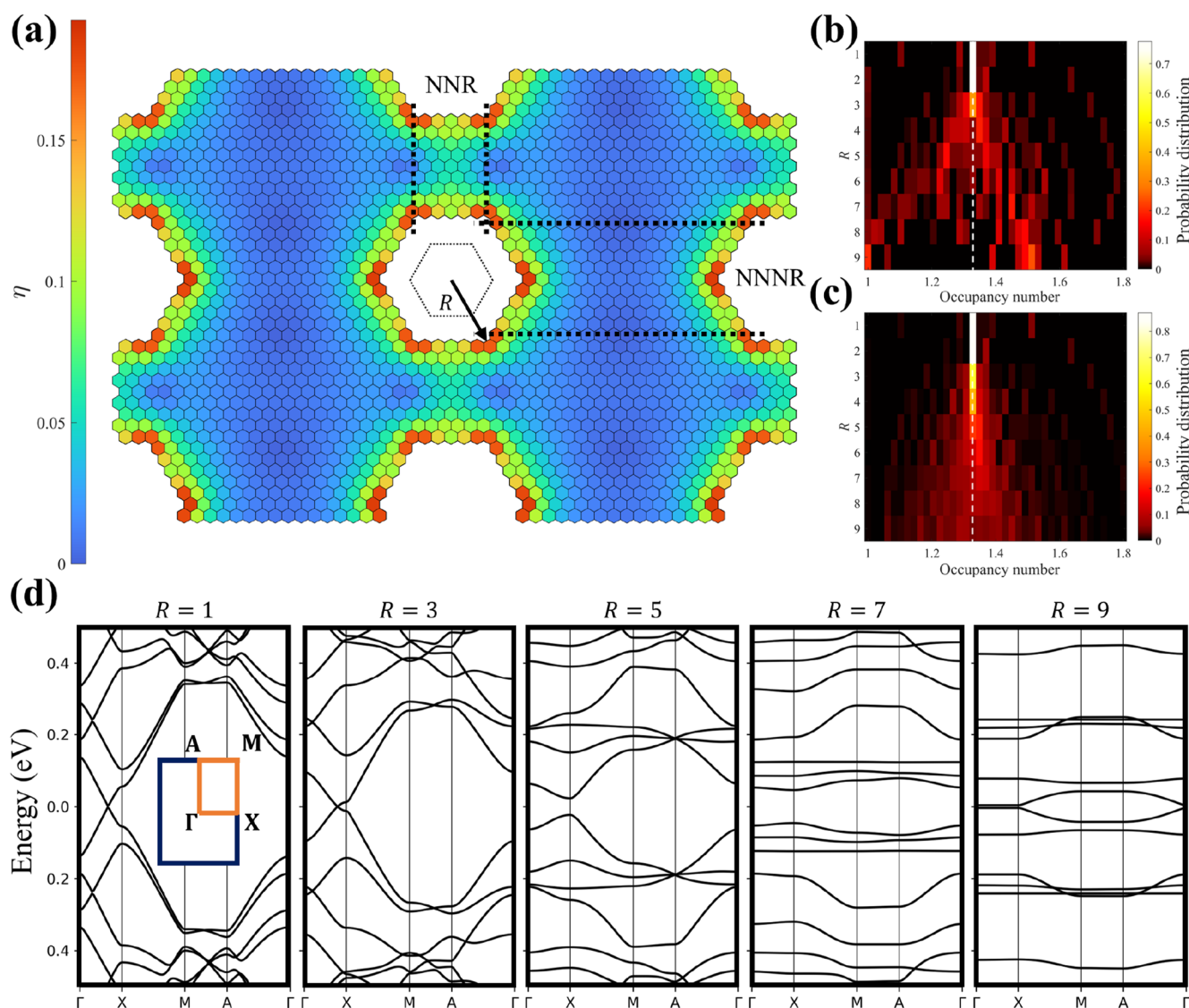


FIG. 6. (a) Schematic of $\{[26, 0], [-10, 20]\}$ with $R = 5$. The regions between the defect and its first and second nearest neighbors are, respectively, delineated with the black dashed lines. (b) and (c) Evolution of ON distribution within NNR and NNNR as R increases. (d) Band structures of $\{[26, 0], [-10, 20]\}$ with increasing R .

with the shape of the defects, leads to a significant anisotropy in the degree of electron localization in rectangular GALs.

To quantify the level of electron delocalization, we introduce a descriptor for local hexagonal rings. In particular, by averaging the deviations of the ONs on the carbon hexagonal rings, the localization parameter η is defined as

$$\eta = \sum_{i=1}^6 \frac{|\text{ON}_i - u_i|}{6}, \quad (4)$$

where i traverses all bonds belonging to the hexagonal ring. Here, u_i is the occupancy number corresponding to a perfectly uniform electron distribution. It means that u_i is generally $\frac{4}{3}$, except for the C–C bonds adjacent to C–H bonds, where u_i is $\frac{3}{2}$. Smaller η indicate a more uniform distribution of electron density and bond lengths, as well as greater delocalization of π -electrons. As examples, we plotted the η distribution for $\{[3,0], \mathbf{A}_2\}$ and $\{[2,2], \mathbf{A}_2\}$ in the [supplementary material](#), which is in agreement with the electron localization function (ELF)⁴³ values.

Figure 6(a) illustrates the distribution of η for $\{[26,0], [-10,20]\}$ with $R = 5$. The first nearest-neighbor region (NNR) and the next nearest-neighbor region (NNNR) based on the distances between the defects in the GALs are indicated by the white dashed lines, respectively. We observed that electron localization primarily occurs at the edges of the defects. In addition, compared to NNNR, η is generally larger in the NNR, which shows more inhomogeneous electron distribution.

As R increases, the occupancy number distributions of $\{[26,0], [-10,20]\}$ in the NNR and NNNR are shown in Figs. 6(b) and 6(c), respectively, with the corresponding changes in band structures shown in Fig. 6(d). It can be seen that in the NNR, the ONs are concentrated around $\frac{4}{3}$ when R is small and then a significant split appears around $R = 5$. Eventually, at $R = 9$, the occupancy numbers on most bonds are distributed near 1 and $3/2$. However, for the C–C bonds in the NNNR, the occupancy number distribution does not show a clear split as depicted in Fig. 6(c), with the sharp peak at $4/3$ gradually becoming more flattened. Combining the analysis of the band structure, Fig. 6(d) shows that the bandgap initially closes along $\Gamma \rightarrow X$. As R increases, the Dirac point gradually shifts toward the high-symmetry point X located at $(1/2, 0)$. At $R = 5$, a bandgap of a certain value opens at X, corresponding to the peak of occupancy number distribution at $4/3$ splits in the NNR. The bandgap reaches its maximum value of ~ 0.1 eV at $R = 7$ and then drops to a very small value due to the excessively long zigzag edge. For GALs such as $\{[26,0], [-10,20]\}$ that do not follow the typical bandgap opening rules, the changes in their band structures can be explained by the anisotropy of the π -electron localization degree. In rectangular GALs, due to the low symmetry of unit cells,

the localization degree of π -electron probability distribution in the NNR increases more rapidly as R increases. This leads to a weakening of interactions along the zigzag direction, which is manifested in Fig. 6(d) as the bands along $\Gamma \rightarrow X$ flattening more quickly than in other regions of reciprocal space. This unique property also explains why $\{[3,0], \mathbf{A}_2\}$ and $\{[2,2], \mathbf{A}_2\}$ exhibit GNR-like bandgap opening patterns in ZH₆-GALs_{8–36}.

After understanding the bandgap opening mechanism dominated by occupancy number distribution, we performed METB calculations on ZH-GALs with different defect radii, aiming to quantitatively investigate the relationship between the bandgap E_g and the structural parameters of rectangular GALs. We selected rectangular ZH-GALs $\{[l,0], [-m,2m]\}$ with $l \geq 2m - 1$ to ensure that defects aggregate preferentially in the armchair direction as R increases. It is obvious that when the defect rate increases, Eq. (3) fails to describe the bandgap variation represented by data points in Fig. 6. In fact, owing to the coexistence of defect scattering and electron localization effects, the bandgap behavior is complex in rectangular ZH-GALs. As electron localization occurs horizontally, 1D channels will gradually form in the perpendicular direction, as shown in Fig. 6(a). Therefore, our results support a scaling law with a period of three. When R is small relative to the lattice vectors, $\{[3n,0], [-m,2m]\}$ open a bandgap, while $\{[3n+1,0], [-m,2m]\}$ and $\{[3n+2,0], [-m,2m]\}$ do not, which is consistent with the typical rule. However, as R increases, $\{[3n+2,0], [-m,2m]\}$ begins to open a bandgap before $\{[3n+1,0], [-m,2m]\}$. In addition, we observed that for structures with larger N_{total} , the bandgap first increases and then decreases with increasing R , consistent with previous findings.²⁸ Thus, we propose a simple scaling law with six parameters for rectangular ZH-GALs,

$$E_g = a \frac{N_{\text{removed}}^b}{N_{\text{total}}^c} \exp\left(-d \frac{N_{\text{removed}}^e}{N_{\text{total}}^f}\right). \quad (5)$$

We use the remainder of $\mathbf{A}_1$ divided by three to categorize them and fit each category with separate parameters, as shown in and Table II. For $\{[3n,0], [-m,2m]\}$ shown in Fig. 7(d), the fitting parameters b and c are 0.56 and 1.02, respectively, allowing our scaling law to revert to Eq. (3) when N_{removed} is small. Figures 7(d)–7(f) shows that for rectangular ZH-GALs with the same remainder, the fitted bandgap peak shifts to larger R as the unit cell volume increases. This indicates that the mechanism causing bandgap differences is no longer the defect scattering described by the typical bandgap opening rule but the π -electron localization due to smaller inter-defect distances. Our scaling law reveals that for structures exhibiting quantum confinement effects in only one dimension, the bandgap remains highly sensitive to the unit cell geometric parameters. In addition, we suppose that the excessively

TABLE II. Numerical fitting results for different data with Eq. (5).

Unit cell structures	a	b	c	d	e	f	RMSE (eV)
$\{[3n,0], [-m,2m]\}$	19.1267	0.5615	1.0229	0.0668	1.6941	0.9263	0.0317
$\{[3n+1,0], [-m,2m]\}$	3455.4955	13.4956	12.4818	3.1659	2.4008	2.0082	0.0225
$\{[3n+2,0], [-m,2m]\}$	38.8999	1.7508	2.2082	0.0001	3.7582	1.7580	0.0205

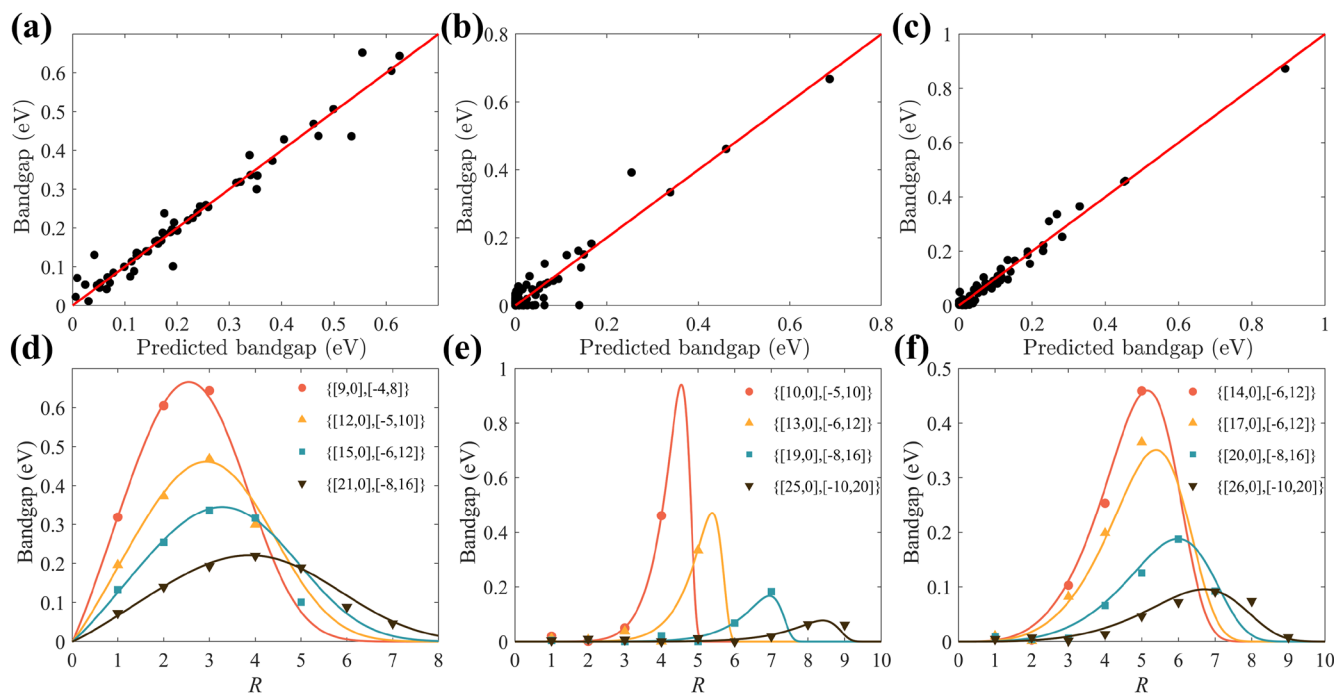


FIG. 7. (a)–(c) Comparison between bandgaps of rectangular ZH-GALs $\{[l, 0], [-m, 2m]\}$ from METB calculations and Eq. (5). (d)–(f) Bandgaps as a function of R for rectangular ZH-GALs $\{[l, 0], [-m, 2m]\}$ with $l = 3n$, $l = 3n + 1$, and $l = 3n + 2$. The solid lines are obtained from Eq. (5), while the data points are derived using the METB method.

long zigzag edges of antidots prevent $\{[3n + 1, 0], [-m, 2m]\}$ and $\{[3n + 2, 0], [-m, 2m]\}$ from achieving a large bandgap through quantum confinement when N_{removed} becomes large, which is significantly different from previous work.^{10,21} Our proposed scaling law provides a good estimation of the bandgap for rectangular ZH-GALs with the least number of parameters.

IV. CONCLUSIONS

In conclusion, we have systematically investigated the electronic properties of ZH-GALs using the maximum entropy method, revealing a complex bandgap opening mechanism induced by π -electron localization between antidots. Due to the low symmetry of unit cell, the anisotropy of electron localization suggests a transition from GALs-like to GNRs-like bandgap behavior. Structures belonging to $\{[3, 0], A_2\}$ and $\{[2, 2], A_2\}$ with $R = 1$ demonstrate behavior that differs from the typical bandgap opening rule. Among them, $\{[3, 0], [-1, 5]\}$ has been experimentally synthesized. Analysis of rectangular ZH-GALs with varying R showed a distinct correlation between bandgap variations and unit cell size changes with a period of three, and the bandgap opening is less restrictive than the typical rule. The scaling law for rectangular ZH-GALs based on METB effectively estimates the bandgap with minimal parameters, supporting the design of GALs with tailored electronic properties. It is shown that the proportion of ZH-GALs with a significant bandgap is actually more than 1/9. Our study, combining electron

density distribution and band structure analysis, provides strong theoretical support for GALs' design and application. It compensates for the lack in band structure engineering between regimes dominated by defect scattering and quantum confinement.

SUPPLEMENTARY MATERIAL

The [supplementary material](#) contains information of the determination of bonding environment parameters, the calculation results of ZH₆-GALs_{8–36}, the electron localization in GALs, the LDOS of π -electron localized GALs, the disordered GALs, the comparison with the Wiberg bond index from natural bond orbital analysis, the computational efficiency for large systems, and the implications of PBE underestimation on METB.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grant Nos. 12474228 and 12474229) and the Basic and Applied Basic Research Foundation of Guangdong Province (Grants No. 2023A1515110894).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Hai-Wei Luo: Data curation (equal); Formal analysis (equal); Investigation (equal); Visualization (equal); Writing – original draft (equal). **Chang-Chun He:** Methodology (equal). **Yu-Jun Zhao:** Writing – review & editing (equal). **Xiao-Bao Yang:** Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#).

REFERENCES

- ¹A. H. Castro Neto, F. Guinea, N. M. R. Peres, K. S. Novoselov, and A. K. Geim, “The electronic properties of graphene,” *Rev. Mod. Phys.* **81**, 109–162 (2009).
- ²A. K. Geim and K. S. Novoselov, “The rise of graphene,” *Nat. Mater.* **6**, 183–191 (2007).
- ³D. G. Papageorgiou, I. A. Kinloch, and R. J. Young, “Mechanical properties of graphene and graphene-based nanocomposites,” *Prog. Mater. Sci.* **90**, 75–127 (2017).
- ⁴A. A. Balandin, “Thermal properties of graphene and nanostructured carbon materials,” *Nat. Mater.* **10**, 569–581 (2011).
- ⁵R. Balog, B. Jørgensen, L. Nilsson, M. Andersen, E. Rienks, M. Bianchi, M. Fanetti, E. Lægsgaard, A. Baraldi, S. Lizzit, Z. Sljivancanin, F. Besenbacher, B. Hammer, T. G. Pedersen, P. Hofmann, and L. Hornekær, “Bandgap opening in graphene induced by patterned hydrogen adsorption,” *Nat. Mater.* **9**, 315–319 (2010).
- ⁶X. Liu, Z. Zhang, and W. Guo, “Universal rule on chirality-dependent bandgaps in graphene antidot lattices,” *Small* **9**, 1405–1410 (2013).
- ⁷C. Moreno, M. Vilas-Varela, B. Kretz, A. Garcia-Lekue, M. V. Costache, M. Paradinas, M. Panighel, G. Ceballos, S. O. Valenzuela, D. Peña, and A. Mugarza, “Bottom-up synthesis of multifunctional nanoporous graphene,” *Science* **360**, 199–203 (2018).
- ⁸W. Oswald and Z. Wu, “Energy gaps in graphene nanomeshes,” *Phys. Rev. B* **85**, 115431 (2012).
- ⁹M. Vanević, V. M. Stojanović, and M. Kindermann, “Character of electronic states in graphene antidot lattices: Flat bands and spatial localization,” *Phys. Rev. B* **80**, 045410 (2009).
- ¹⁰T. G. Pedersen, C. Flindt, J. Pedersen, N. A. Mortensen, A.-P. Jauho, and K. Pedersen, “Graphene antidot lattices: Designed defects and spin qubits,” *Phys. Rev. Lett.* **100**, 136804 (2008).
- ¹¹Z. Wang, R. Yin, J. Meng, J. Wang, Y. Liang, C. Ma, S. Tan, Q. Li, J. Yang, and B. Wang, “Self-limited embedding alternating 585-ringed divacancies and metal atoms into graphene nanoribbons,” *J. Am. Chem. Soc.* **145**, 8445–8454 (2023).
- ¹²R. Yin, J. Wang, Z.-L. Qiu, J. Meng, H. Xu, Z. Wang, Y. Liang, X.-J. Zhao, C. Ma, Y.-Z. Tan, Q. Li, and B. Wang, “Step-assisted on-surface synthesis of graphene nanoribbons embedded with periodic divacancies,” *J. Am. Chem. Soc.* **144**, 14798–14808 (2022).
- ¹³J. Bai, X. Zhong, S. Jiang, Y. Huang, and X. Duan, “Graphene nanomesh,” *Nat. Nanotechnol.* **5**, 190–194 (2010).
- ¹⁴T. de Sousa Araújo Cassiano, F. F. Monteiro, L. Evaristo de Sousa, G. Magela e Silva, and P. H. de Oliveira Neto, “Smooth gap tuning strategy for cove-type graphene nanoribbons,” *RSC Adv.* **10**, 26937–26943 (2020).
- ¹⁵T.-J. Liu, F. M. Arnold, A. Ghasemifard, Q.-L. Liu, D. Golze, A. Kuc, and T. Heine, “Electronic structure and topology in gulf-edged zigzag graphene nanoribbons,” *Phys. Rev. Mater.* **9**, 014203 (2025).
- ¹⁶Y.-W. Son, M. L. Cohen, and S. G. Louie, “Energy gaps in graphene nanoribbons,” *Phys. Rev. Lett.* **97**, 216803 (2006).
- ¹⁷L. Ci, L. Song, C. Jin, D. Jariwala, D. Wu, Y. Li, A. Srivastava, Z. F. Wang, K. Storr, L. Balicas, F. Liu, and P. M. Ajayan, “Atomic layers of hybridized boron nitride and graphene domains,” *Nat. Mater.* **9**, 430–435 (2010).
- ¹⁸M. T. Lusk and L. D. Carr, “Nanoengineering defect structures on graphene,” *Phys. Rev. Lett.* **100**, 175503 (2008).
- ¹⁹X. Chen, S. Zhang, D. Hou, H. Duan, B. Deng, Z. Zeng, B. Liu, L. Sun, R. Song, J. Du, P. Gao, H. Peng, Z. Liu, and L. Wang, “Tunable pore size from sub-nanometer to a few nanometers in large-area graphene nanoporous atomically thin membranes,” *ACS Appl. Mater. Interfaces* **13**, 29926–29935 (2021).
- ²⁰X. Han, M. R. Funk, F. Shen, Y.-C. Chen, Y. Li, C. J. Campbell, J. Dai, X. Yang, J.-W. Kim, Y. Liao, J. W. Connell, V. Barone, Z. Chen, Y. Lin, and L. Hu, “Scalable holey graphene synthesis and dense electrode fabrication toward high-performance ultracapacitors,” *ACS Nano* **8**, 8255–8265 (2014).
- ²¹B. S. Jessen, L. Gammelgaard, M. R. Thomsen, D. M. A. Mackenzie, J. D. Thomsen, J. M. Caridad, E. Duegaard, K. Watanabe, T. Taniguchi, T. J. Booth, T. G. Pedersen, A.-P. Jauho, and P. Bøggild, “Lithographic band structure engineering of graphene,” *Nat. Nanotechnol.* **14**, 340–346 (2019).
- ²²M. R. Thomsen and T. G. Pedersen, “Analytical Dirac model of graphene rings, dots, and antidots in magnetic fields,” *Phys. Rev. B* **95**, 235427 (2017).
- ²³S. Song, Y. Teng, W. Tang, Z. Xu, Y. He, J. Ruan, T. Kojima, W. Hu, F. J. Giessibl, H. Sakaguchi, S. G. Louie, and J. Lu, “Janus graphene nanoribbons with localized states on a single zigzag edge,” *Nature* **637**, 580–586 (2025).
- ²⁴P. H. Jacobse, R. D. McCurdy, J. Jiang, D. J. Rizzo, G. Veber, P. Butler, R. Zuzak, S. G. Louie, F. R. Fischer, and M. F. Crommie, “Bottom-up assembly of nanoporous graphene with emergent electronic states,” *J. Am. Chem. Soc.* **142**, 13507–13514 (2020).
- ²⁵T. Qin, T. Wang, and J. Zhu, “Recent progress in on-surface synthesis of nanoporous graphene materials,” *Commun. Chem.* **7**, 154 (2024).
- ²⁶M. Dvorak, N. Oswald, and Z. Wu, “Bandgap opening by patterning graphene,” *Sci. Rep.* **3**, 2289 (2013).
- ²⁷M. Dvorak and Z. Wu, “Geometrically induced transitions between semimetal and semiconductor in graphene,” *Phys. Rev. B* **90**, 115415 (2014).
- ²⁸F. Ouyang, S. Peng, Z. Yang, Y. Chen, H. Zou, and X. Xiong, “Bandgap opening/closing of graphene antidot lattices with zigzag-edged hexagonal holes,” *Phys. Chem. Chem. Phys.* **16**, 20524–20531 (2014).
- ²⁹F. Ouyang, Z. Yang, S. Peng, X. Zheng, and X. Xiong, “Antidot-dependent bandgap and Clar sextets in graphene antidot lattices,” *Physica E* **56**, 222–226 (2014).
- ³⁰R. Petersen, T. G. Pedersen, and A.-P. Jauho, “Clar sextet analysis of triangular, rectangular, and honeycomb graphene antidot lattices,” *ACS Nano* **5**, 523–529 (2011).
- ³¹S. S. Gregersen, S. R. Power, and A.-P. Jauho, “Robust bandgap and half-metallicity in graphene with triangular perforations,” *Phys. Rev. B* **93**, 245429 (2016).
- ³²G. Calogero, N. R. Papior, B. Kretz, A. Garcia-Lekue, T. Frederiksen, and M. Brandbyge, “Electron transport in nanoporous graphene: Probing the talbot effect,” *Nano Lett.* **19**, 576–581 (2019).
- ³³C. Moreno, X. Diaz de Cerio, M. Vilas-Varela, M. Tenorio, A. Sarasola, M. Brandbyge, D. Peña, A. Garcia-Lekue, and A. Mugarza, “Molecular bridge engineering for tuning quantum electronic transport and anisotropy in nanoporous graphene,” *J. Am. Chem. Soc.* **145**, 8988–8995 (2023).
- ³⁴I. Piquero-Zulaica, E. Corral-Rascón, X. Diaz de Cerio, A. Riss, B. Yang, A. Garcia-Lekue, M. A. Kher-Elden, Z. M. Abd El-Fattah, S. Nobusue, T. Kojima, K. Seufert, H. Sakaguchi, W. Auwärter, and J. V. Barth, “Deceptive orbital confinement at edges and pores of carbon-based 1D and 2D nanoarchitectures,” *Nat. Commun.* **15**, 1062 (2024).
- ³⁵G. Calogero, I. Alcón, O. Kaya, N. Papior, A. W. Cummings, M. Brandbyge, and S. Roche, “Robust quantum engineering of current flow in carbon nanostructures at room temperature,” *Carbon* **234**, 119950 (2025).

- ³⁶G. Calogero, I. Alcón, N. Papior, A.-P. Jauho, and M. Brandbyge, “Quantum interference engineering of nanoporous graphene for carbon nanocircuitry,” *J. Am. Chem. Soc.* **141**, 13081–13088 (2019).
- ³⁷I. Langmuir, “The octet theory of valence and its applications with special reference to organic nitrogen compounds,” *J. Am. Chem. Soc.* **42**, 274–292 (1920).
- ³⁸C.-C. He, S.-G. Xu, J. Zeng, W. Huang, Y. Yao, Y.-J. Zhao, H. Xu, and X.-B. Yang, “A graph-based statistical model for carbon nanostructures,” *J. Chem. Phys.* **162**, 154104 (2025).
- ³⁹C.-C. He, S.-G. Xu, Y.-J. Zhao, H. Xu, and X.-B. Yang, “Entropy-driven electron density and effective model Hamiltonian for boron systems,” *Phys. Rev. B* **111**, 085408 (2025).
- ⁴⁰G. Kresse and J. Furthmüller, “Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set,” *Phys. Rev. B* **54**, 11169–11186 (1996).
- ⁴¹G. Kresse and J. Furthmüller, “Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set,” *Comput. Mater. Sci.* **6**, 15–50 (1996).
- ⁴²J. P. Perdew, K. Burke, and M. Ernzerhof, “Generalized gradient approximation made simple,” *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
- ⁴³B. Silvi and A. Savin, “Classification of chemical bonds based on topological analysis of electron localization functions,” *Nature* **371**, 683–686 (1994).